

VOLVO PENTA	Document No	Issue Index
TAD843GE	24132160	01

Important

This Technical Data Sheet and the corresponding Installation Instructions provide important information to ensure the installed engine will operate according to the design specification in the Volvo Penta application for certification.

Requirements marked with  are considered as critical for exhaust emissions compliance according to the design specification in the Volvo Penta application for certification.

Failing to follow and meet these instructions and requirements when installing a certified engine in a piece of nonroad equipment for use in the United States violates U.S. federal law (40 CFR 1068.105(b)), subject to fines or other penalties as described in the Clean Air Act.

General

In-line four stroke turbocharged diesel engine with direct injection.

Rotation direction, anti-clockwise viewed towards flywheel.

Number of cylinders			6
Displacement, total		litre in ³	7,70 469,9
Firing order			1-4-2-6-3-5
Bore		mm in	110 4,33
Stroke		mm in	135 5,31
Compression ratio			17.5:1
Wet weight w/o EATS	Engine only	kg lb	707 1559
	Engine incl. cooling system and air filtration system	kg lb	917 2022
	Engine incl. cooling system, air filtration system, and frame	kg lb	N/A

Performance

Performance		rpm	1500	1800
Standby Power	without fan	kW	320	321
		hp	435	437
	with fan	kW	308	301
		hp	419	409
Prime Power	without fan	kW	292	294
		hp	397	400
	with fan	kW	280	274
		hp	381	373
COP Power	without fan	kW	219	221
		hp	298	300
	with fan	kW	207	201
		hp	282	273
Torque at:	Standby Power	Nm	2037	1703
		lbft	1502	1256
	Maximum within fine speed range	Nm	2167	1793
		lbft	1598	1322
Total mass moment of inertia, J (mR ²)		kgm ² lbft ²	0,420 10,0	
Derating due to altitude - see Technical Diagrams				

VOLVO PENTA TAD843GE	Document No 24132160	Issue Index 01
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Engine noise emission

Test Standards: ISO 3744-1981 (E) sound power with fan

Tolerance ± 0.75 dB(A)

		rpm	1500	1800
Measured sound power Lw	Standby Power	dB(A)	111,8	115,3
	Prime Power	dB(A)	111,2	115,3
	No load	dB(A)	111,5	100,3
Calculated sound pressure Lp at 1 m	Standby Power	dB(A)	99,8	103,3
	Prime Power	dB(A)	99,2	103,3
	No load	dB(A)	99,5	88,3

Tailpipe noise

Data calculated as sound pressure Lp.

Assumed microphone distance 1 m?

	rpm	1500	1800
Standby Power	dB(A)	N/A	N/A
Prime Power	dB(A)	N/A	N/A
COP	dB(A)	N/A	N/A

Test conditions for load acceptance data

Engine at working temperature, fuel that is used..... Nominal operating conditions

Generator	Brand	Model	Type of AVR
AVR Settings	UFRO (Hz):	DIP:	DWELL:
	Stability (%)*:	Voltage (V):	Power factor:

Load acceptance performance can vary due to actual alternator inertia, voltage regulator, type of load and local ambient conditions.

Nomenclature

Abbreviation:	Full name:	Descriptions
AVR	Automatic Voltage Regulator	Generator performance and safety control unit
UFRO	Under Frequency Roll Off	Overheating protection at under frequency
-	Dip	Controls the slope of voltage drop when the UFRO is active
-	Dwell	Controls the slope of voltage recovery when the UFRO is active.

Load Acceptance at 1500 rpm

Genset Classification

This engine fulfills G1, G2 and G3 classes, according to ISO8528-5. For other class, please, see the table below.

G4			
Load (%)	Speed diff (%)	Speed Recovery time (s)	
0-39	7 (G3)	2,0	G3 boundary conditions
0-48	10 (G2)	3,0	G2 boundary conditions

Load (%)	Speed diff (%)	Speed Recovery time (s)	Voltage diff (%)	Voltage Recovery time (s)	Remaining load (%)	Speed diff (%)	Speed Recovery time (s)	Voltage diff (%)	Voltage Recovery time (s)
0-20	3,0	1,2	1,1	0,2	20-100	25,7	5,8	54,6	4,8
0-40	7,6	2,4	3,1	1,8	40-100	10,1	2,4	7,2	1,5
0-60	22,1	3,0	31,0	2,5	60-100	3,2	1,2	3,7	0,4
0-80	50,3	6,5	56,6	5,6	80-100	1,1	1,0	0,8	0,1
0-100	N/A	N/A	N/A	N/A					
0-110	N/A	N/A	N/A	N/A					
100-0	13,8	1,9	12,8	1,3					

VOLVO PENTA	Document No	Issue Index
TAD843GE	24132160	01

Load Acceptance at 1800 rpm

Genset Classification
This engine fulfills G1, G2 and G3 classes, according to ISO8528-5. For other class, please, see the table below.

Load (%)	Speed diff (%)	Speed Recovery time (s)	
0-55	7 (G3)	1,5	G3 boundary conditions
0-59	10 (G2)	1,8	G2 boundary conditions

Load (%)	Speed diff (%)	Speed Recovery time (s)	Voltage diff (%)	Voltage Recovery time (s)	Remaining load (%)	Speed diff (%)	Speed Recovery time (s)	Voltage diff (%)	Voltage Recovery time (s)
0-20	1,5	0,6	0,7	0,2	20-100	13,3	2,2	16,9	2,0
0-40	3,6	0,8	1,7	1,1	40-100	6,1	1,2	6,1	1,2
0-60	8,7	1,8	4,3	1,9	60-100	3,8	0,9	4,3	0,8
0-80	18,1	2,5	25,4	2,4	80-100	1,6	0,5	2,2	0,4
0-100	N/A	N/A	N/A	N/A					
0-110	N/A	N/A	N/A	N/A					
100-0	6,8	1,3	6,1	0,4					

Cold start performance	Ambient Temp. [°C]	Manifold Heater	Block heater	RPM	
				1500	1800
Time to Set Speed from start	20	-	-	4,8	5,5
	5	-	-	4,3	5,5
	-15*	Yes	-	6,4	7,1
	-25*	Yes	-	8,6	-
	-30 **	Yes	Yes	5,3	7,9

Min start temp w/o Block Heater*	-25	°C
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* With manifold heater kW engaged, lubrication oil SAE 10W/30.

** With manifold heater kW engaged, lubrication oil SAE 10W/30 and block heater, Fuel MK-1.

Block heater type	Power kW	Engaged hours	Cooling water temp engine block
M9T701	1,5	16	28°C

VOLVO PENTA	Document No	Issue Index
TAD843GE	24132160	01

Lubrication system			rpm	1500	1800
Lubricating oil consumption	Standby Power		litre/h	0,02	0,02
			US gal/h	0,005	0,005
	Prime Power		litre/h	0,02	0,02
			US gal/h	0,005	0,005
	COP		litre/h	N/A	N/A
			US gal/h		
Oil system capacity including filters			litre	27	
			US gal	7,1	
Oil sump capacity:		max	litre	25	
			US gal	6,6	
		min	litre	16	
			US gal	4,2	
Oil change intervals/specifications:			h	500	
Engine angularity limits:		front up	°	10	
		front down	°	10	
		side tilt	°	10	
Oil pressure at nominal set speed			kPa	330 - 430	
			psi	48 - 62	
Lubrication oil temperature in oil sump:		max	°C	125	
			°F	257	
Oil filter micron size			μ	5,000	

* See also general section in the sales guide

VOLVO PENTA	Document No	Issue Index
TAD843GE	24132160	01

Intake and exhaust system		rpm	1500	1800
Air consumption at: (+25°C and 100kPa)	Standby Power	m ³ /min cfm	19 671	21,8 770
	Prime Power	m ³ /min cfm	18,7 660	21,9 773



See front page for important information

Max air intake restriction including piping with maintained performance		kPa psi	3 0,4	3,8 0,6
Max <u>allowable</u> air intake restriction including piping		kPa psi	5 0,7	5 0,7
Air filter restriction clean Volvo Penta filter		kPa psi	5,0 0,7	5,0 0,7
Heat rejection to exhaust at:	Standby Power	kW BTU/min		
	Prime Power	kW BTU/min		
Exhaust gas temperature after turbine at:	Standby Power	°C	455	500
		°F	851	932
	Prime Power	°C	500	460
		°F	932	860



See front page for important information

Max allowable back pressure in exhaust after turbine	kPa psi	10 1,5	10 1,5
 See front page for important information Max allowable temperature drop between turbine and SCR muffler inlet.			
	Δ°C Δ°F	N/A	N/A
Heat rejection to exhaust:	kW BTU/min		
Exhaust gas temperature after turbine at maximum power:	°C °F	505 941	459 858
 See front page for important information Max allowable temperature drop between turbine and muffler 1 inlet at exhaust temperature 480° C and exhaust gas flow 69.8 m ³ /min.	Δ°C Δ°F	N/A	N/A
 See front page for important information Max allowable temperature drop between muffler 1 and muffler 2 at exhaust temperature ° C and exhaust gas flow m ³ /min.	Δ°C Δ°F	N/A	N/A
DPF muffler pressure drop (at exhaust gas flow and exhaust temp specified in this table)	kPa psi	N/A	N/A
SCR muffler pressure drop (at exhaust gas flow and exhaust temp specified in this table)	kPa psi	N/A	N/A
Exhaust gas flow at max power: (temp and pressure after turbine)	m ³ /min cfm		
 See front page for important information Engine speed during service regeneration			
	rpm	N/A	N/A

 See front page for important information Max allowed load during service regeneration	Nm lb ft	N/A	N/A
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VOLVO PENTA	Document No	Issue Index
TAD843GE	24132160	01

Charge air cooler system		rpm	1500	1800
Heat rejection to charge air cooler at standby power		kW	70	81,3
		BTU/min	3981	4623
Charge air mass flow at standby power		kg/s	0,37	0,43
Charge air inlet temp at standby power (Charge air temp after turbo compressor)	°C	230	226	
	°F	446	439	
 See front page for important information				
Max allowable Charge air outlet temp at standby power (Charge air temp after intercooler)	°C	45	45	
	°F	113	113	
 Maximum pressure drop over charge air cooler incl. Piping		kPa	9	12
		psi	1,31	1,74
Maximum charge air pressure (After charge air cooler)	kPa	2,9	2,7	
	psi	0,42	0,39	
Standard charge air cooler core area	m²	0,217		
	foot²	2,34		

VOLVO PENTA	Document No	Issue Index
TAD843GE	24132160	01

Cooling system

Coolant type and mixture		VCS 40/60		
Coolant capacity,	engine only	litre	17	
		US gal	4,49	
	charge air coolers	litre	N/A	
		US gal		
	coolant radiators incl piping	litre	19	
		US gal	5,02	
	expansion tank	litre	5	
		US gal	1,32	
		rpm	1500	1800
Heat rejection radiation from engine at Standby power:		kW		
		BTU/min		
Heat rejection to coolant at standby power		kW		
		BTU/min		
Standard radiator core area		m²	0,485	
		foot²	5,22	
Min coolant flow engine coolant circuit (at fully open thermostat)		litre/s	3,8	4,6
		US gal/s	1,00	1,22
Maximum coolant temperature entering engine (25°C amb. Temp.)		°C	-	
		F		
Maximum external engine coolant circuit restriction, including piping (25°C amb. Temp.)		kPa	25	
		psi	3,6	
Nominal coolant pressure		kPa	75	75
		psi	10,9	10,9
Nominal coolant flow with standard system		litre/s	3,8	4,6
		US gal/s	1,00	1,22
Fan diameter		mm	650	
		in	25,59	
Fan power consumption		kW	12	20
		hp	16	27
Fan drive ratio			1.4:1	
Coolant pump		drive/ratio	1.4:1	
Thermostat	start to open	°C	85	
		°F	185	
	fully open	°C	100	
		°F	212	
Maximum static pressure head (expansion tank height + pressure cap setting)		kPa	110	
		psi	16,0	
Minimum static pressure head (expansion tank height + pressure cap setting)		kPa	60	
		psi	8,7	
Standard pressure cap setting		kPa	100	
		psi	14,5	
Maximum top tank temperature		°C	107	
		°F	225	
Charge air pressure (after charge air coolers)		kPa	280	
		psi	40,6	

Cooling performance

Standard fan:	650mm	Fan ratio:	1	Fan type:	Fixed
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Cooling air flow and external restriction at different radiator air temperatures based on X°C TTT and X% antifreeze. Valid at 1 atm.
(radiator and cooling fan, see optional equipment)

Engine speed rpm	External restriction Pa	Air flow m ³ /s	STANDBY POWER	PRIME POWER
			Air on temp °C	Air on temp °C
1500	0	6,2	58	59,5
	100			
	200			
	300			
1800	0	7,4	61	62
	100			
	200			
	300			

Note! External restrictions are calculated for values >0 Pa

TAD843GE
Engine management system

Functionality	Alternatives	Default setting
Governor mode	Isochronous / Droop	Isochronous
Governor droop	0-8%	0,0
Governor response	Adjustable PID-constant (VODIA)	Standard
Dual speed	YES	1500 / 1800
Idle speed	600-1200	900,0
Fine speed adjustment	± 90	0
Stop function	Energized to Run / Stop	Energized to Run / Stop
Preheating function	On / Off	On

Engine protection map

Parameter		Unit	Warning Level (Yellow)		Engine protection			
					Alarm level (Red)	Default		Optional
Oil temp		°C		125	130	Shut Down		
Oil pressure	Low idle	kPa		151	101	Shut Down		
	1500 rpm	kPa		233	183	Shut Down		
	1800 rpm	kPa		263	213	Shut Down		
Oil level				N/A	N/A	N/A		
DEF Dosing injector failure				N/A	N/A	N/A		
				N/A	N/A	N/A		
Piston cooling pressure >1000 rpm		kPa		N/A	N/A	N/A		
Coolant temp		°C		105	107,0	Shut Down		
Coolant level				N/A	Low	Shut Down (10 s delay)		
Fuel feed pressure	Low idle	kPa	N/A	N/A	N/A	N/A	N/A	N/A
	>1400 rpm		N/A	N/A	N/A	N/A	N/A	N/A
Water in fuel				On	N/A	N/A		
Crank case pressure		kPa		N/A	N/A	N/A		
Air filter pressure droop		kPa		5,0	N/A	Warning		
Altitude, above sea		m						
Charge air temp		°C		80	85,0	Shut Down		
Charge air pressure		kPa		95-330	200-435	Shut Down		
Engine speed		rpm						
Exhaust Temperature (Before SCR volume)		°C		N/A	N/A	N/A		
				N/A	N/A	N/A		

VOLVO PENTA	Document No	Issue Index
TAD843GE	24132160	01

Electrical system

Voltage and type		24V Dc	
Alternator:	make/output	A	Mitshubishi / 110 A
	tacho output	Hz/alt. Rev	
	drive ratio		1:4
Starter motor	make		SEG
	type		HXF88
	kW		5,6
Number of teeth on:	flywheel		137
	starter motor		10
Max wiring resistance main circuit		mΩ	5
Cranking current at +20°C		A	507
Crank engine speed at 20°C		rpm	230
Starter motor battery capacity:	min	Ah	100 / 680
	CCA at -18°C	Ah/A	140 / 800
Inlet manifold heater (at 24 V)		kW	
Power relay for the manifold heater		A	200









